Olist Delivery Time Prediction

Presented by:

Nouf Almutaweaa & Wadeaa Almutaweaa

Agenda



01/

About Almutaweaa Insights



02/

Problem Statment & Objectives



03/

Data Preprocessing



04/

Model Building and Selection



05/

Live Demo & Conclusion

About Us

Almutaweaa Insights is a **data analytics company** that helps businesses make better decisions through data analysis, predictive modeling, and **machine learning**. We transform data into valuable insights and develop solutions that improve business performance, optimize operations, and support more accurate forecasting.

Almutaweaa
INSIGHTS
COMPANY DATA ANALYTICS

Problem Statment

An analysis of **96,470 completed orders** revealed that **91.8%** arrived **earlier** than estimated, while **6.8%** arrived **late** and only **1.3%** arrived **on time**. This significant gap suggests that Olist's delivery estimates are often overly conservative and do not fully reflect its actual delivery performance, potentially affecting customer expectations and highlighting the need for more accurate estimates.

Objectives

1. **Understand Delivery Performance:** Identify factors affecting delivery times.
2. **Analyze Historical Data:** Prepare reliable data for prediction.
3. **Develop a Prediction Model:** Estimate delivery dates more accurately.
4. **Evaluate Accuracy:** Compare model results with actual deliveries.
5. **Test the Solution:** Demonstrate the model in a shopping application.
6. **Provide Recommendations:** Suggest improvements to delivery estimates.

Understand Olist Delivery Process

Step1: Customer makes an order

Step2: Order gets confirmed

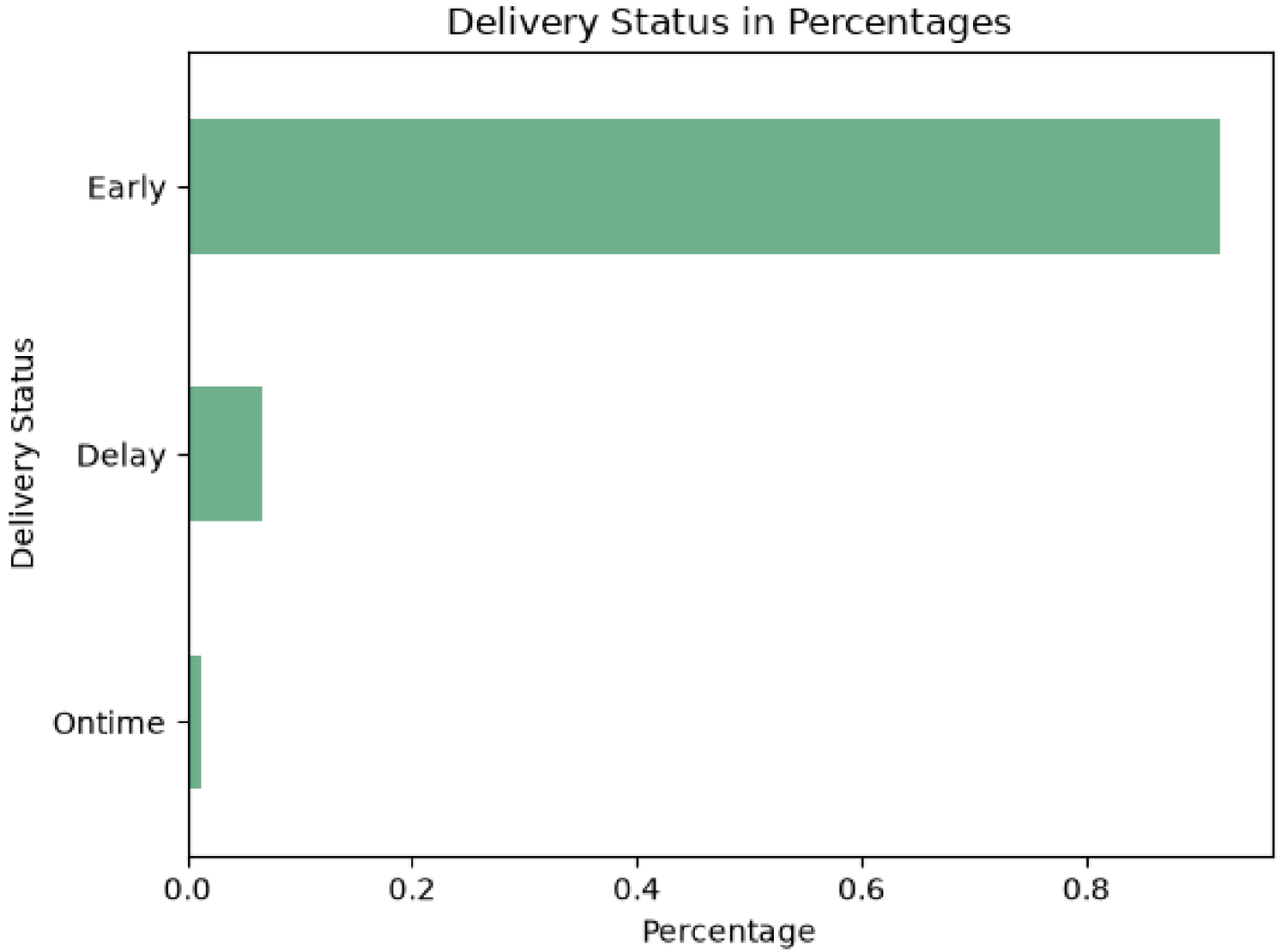
Step3: Olist provides estimated delivery dates

Step4: Shipping company collects items from sellers, and dates are recorded in the system

step5: Shipping company delivers to customer



Delivery Status Overview

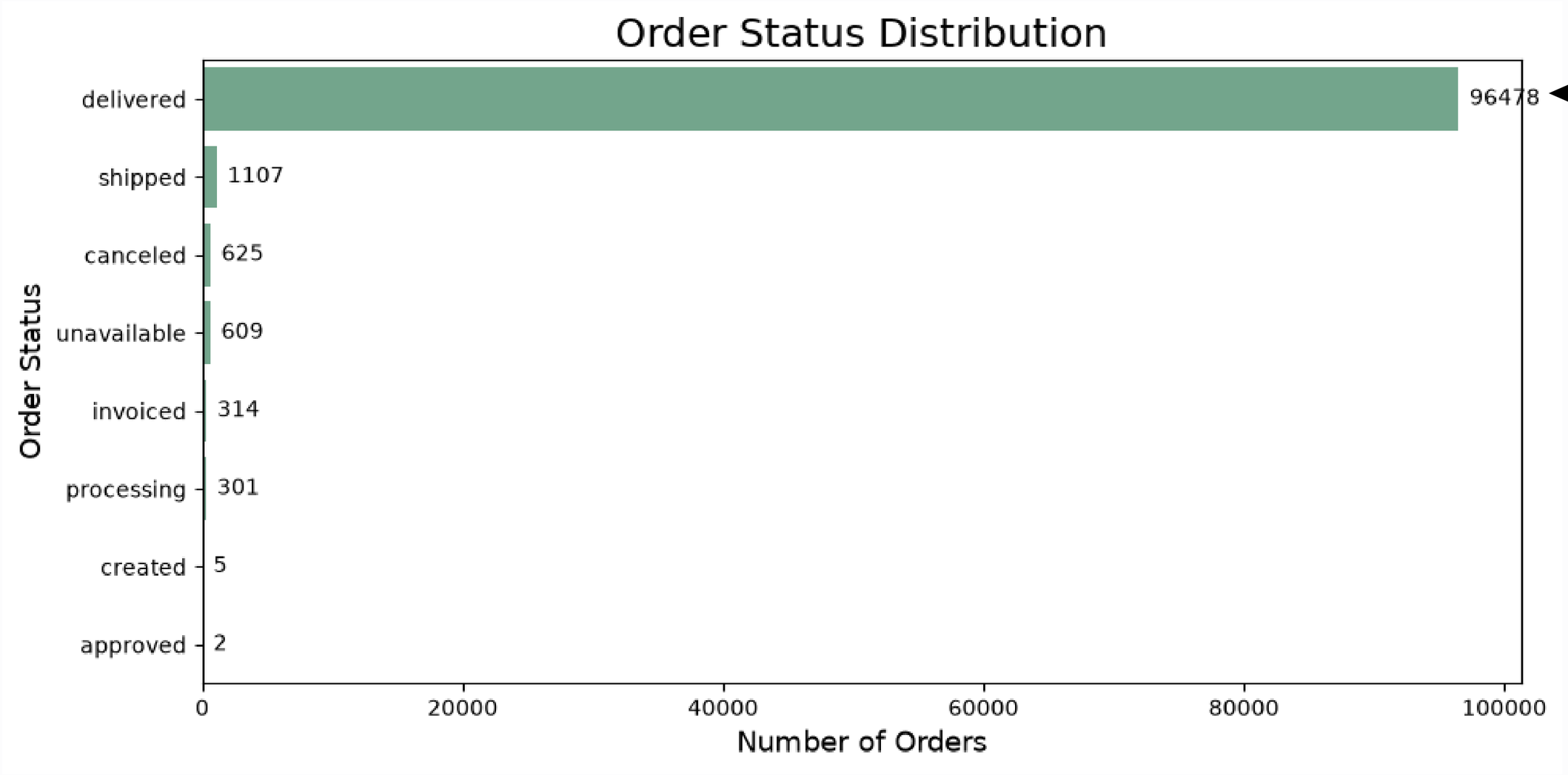


Early: 91.8%

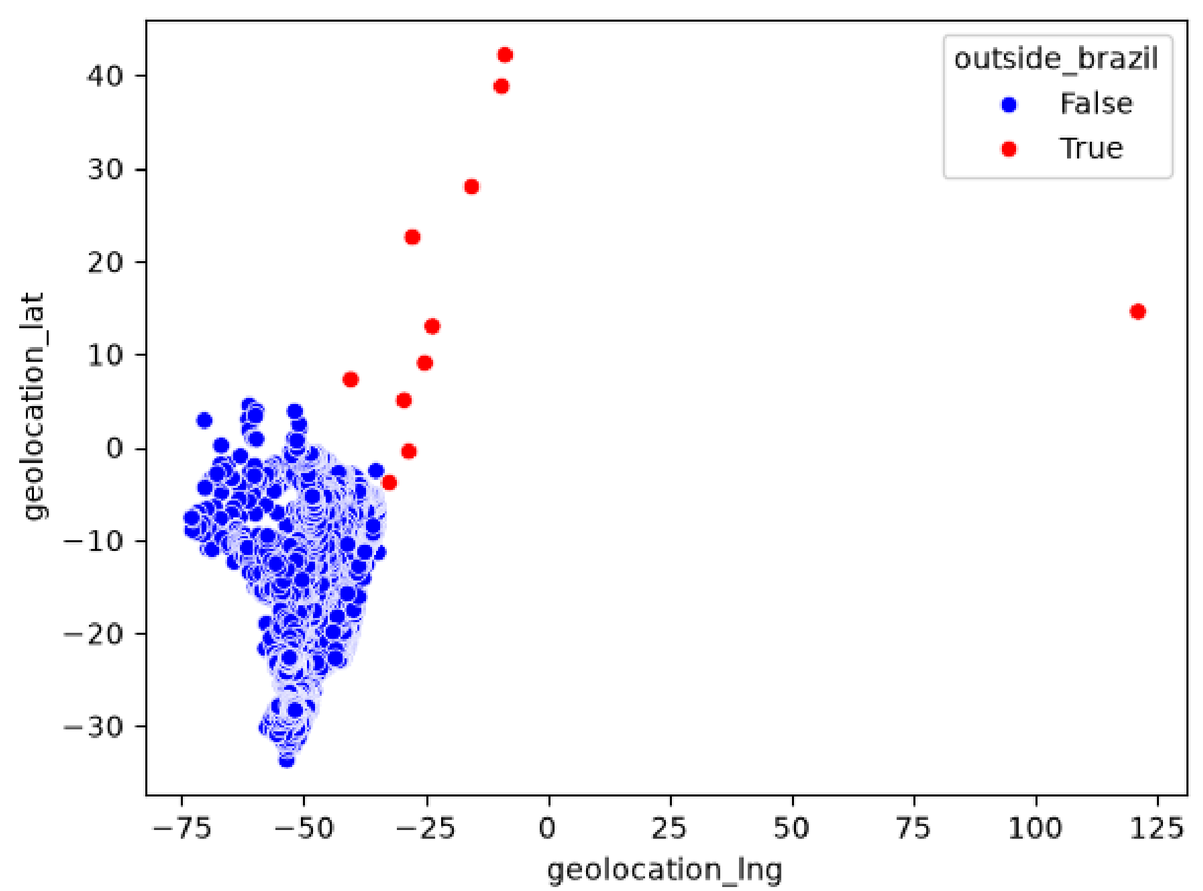
Delay: 6.8%

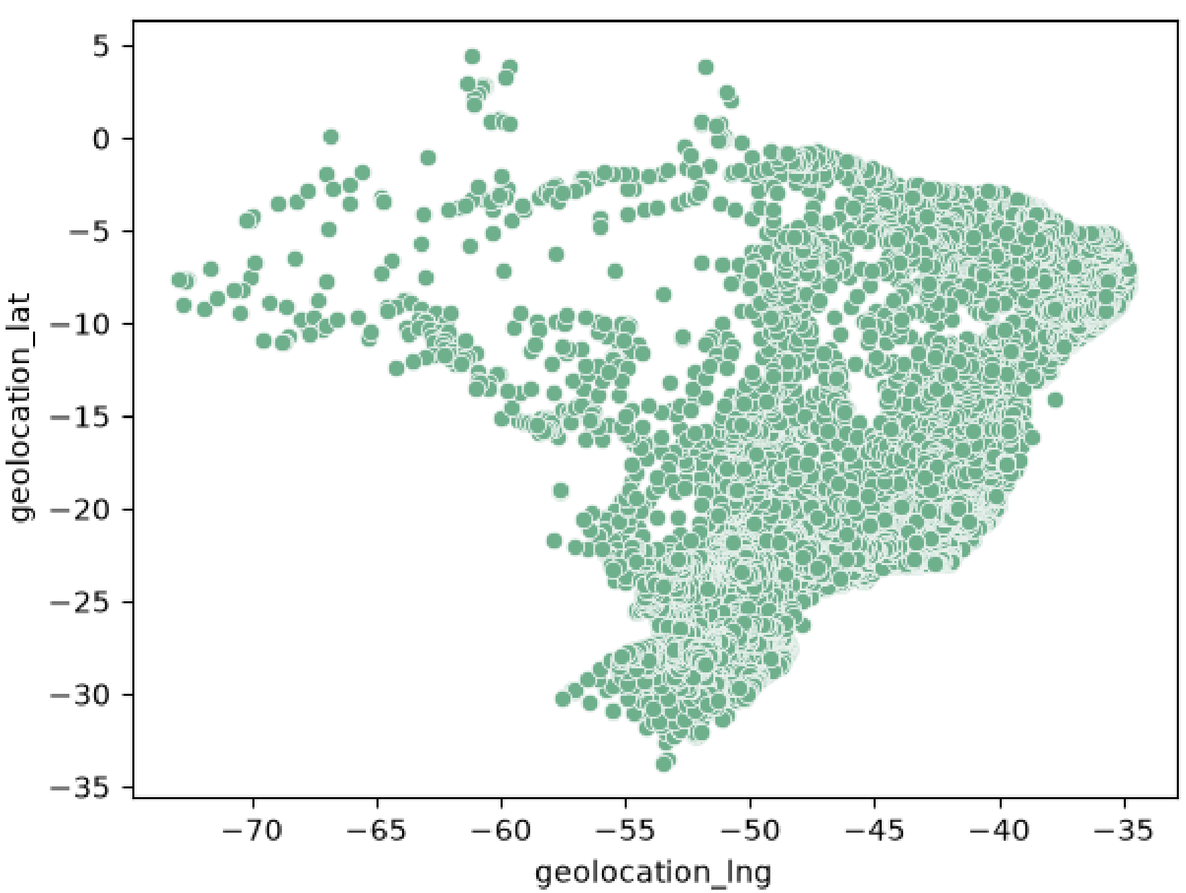
On Time: 1.3%

Data Preprocessing

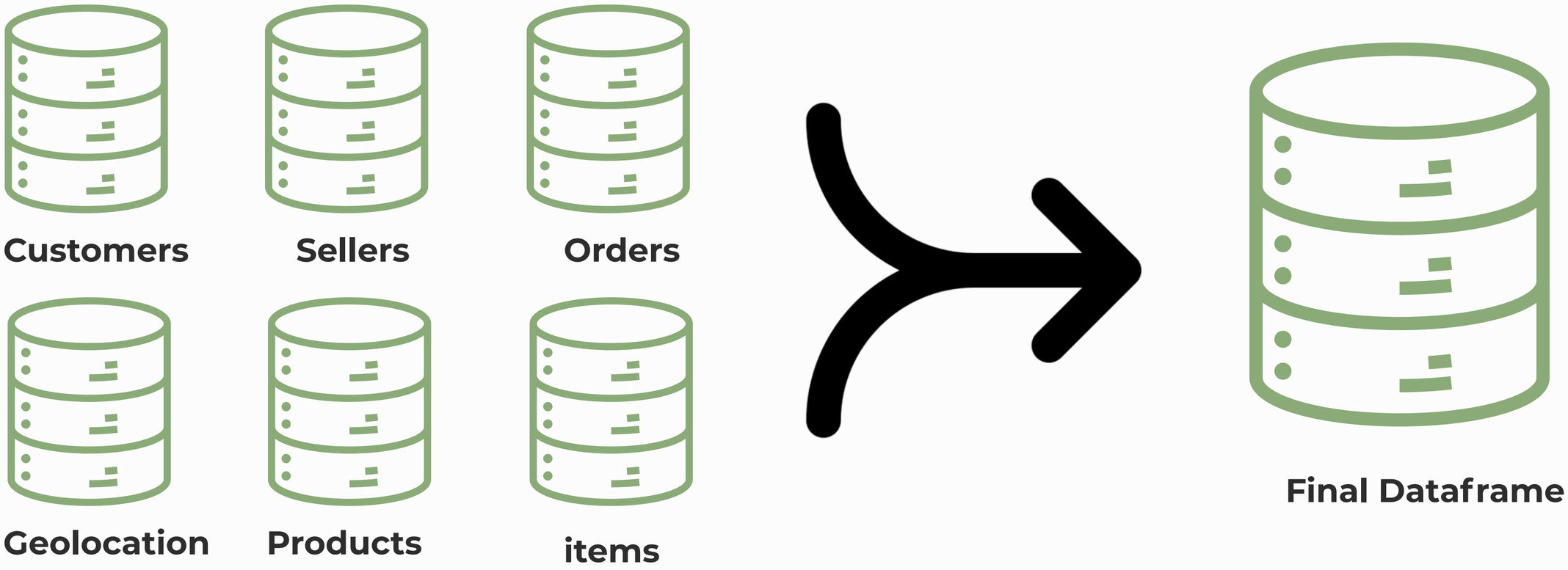


Data Preprocessing





DataSet Overview

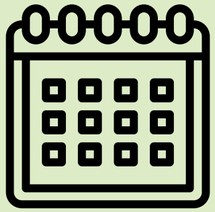


What Does the Final Dataset Contains?



Order Details

Number of sellers and items, product size, and volume



Purchase info

Purchase month and day



Financial Details

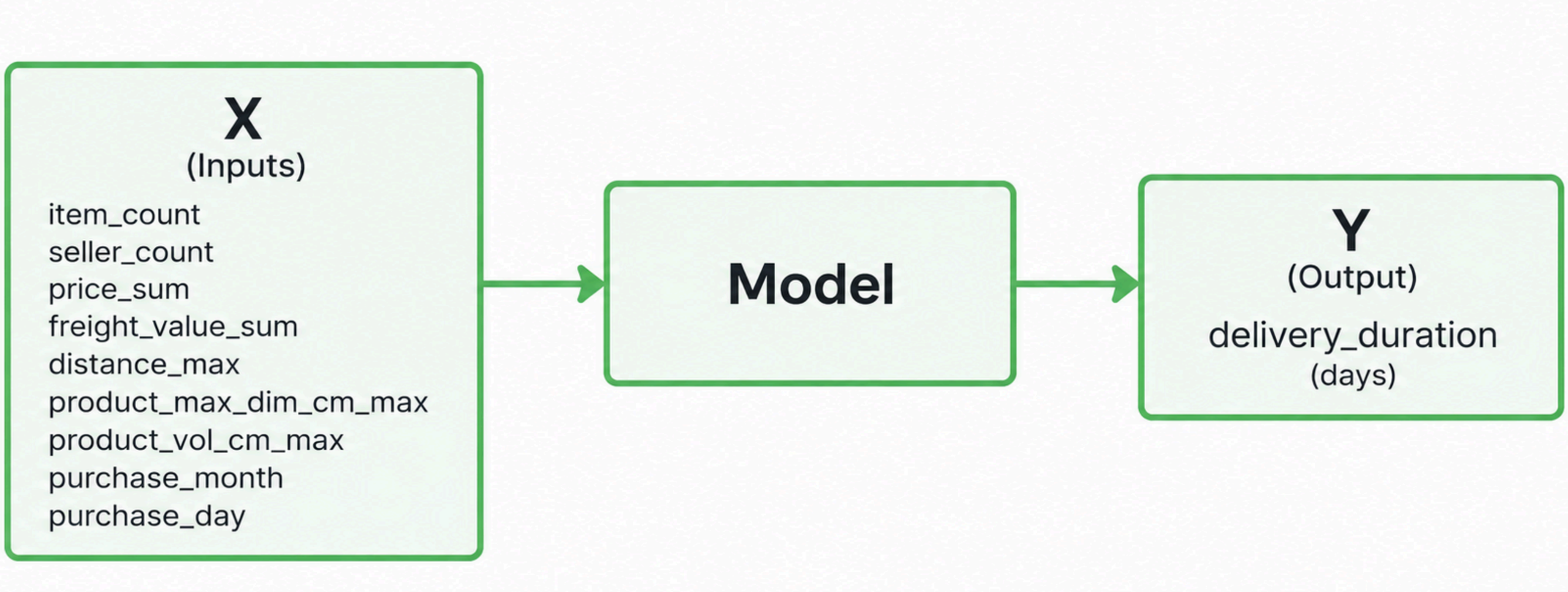
Total order price, and Shipping cost



Location Info

seller-to-customer distance in KM

Delivery Estimation Model



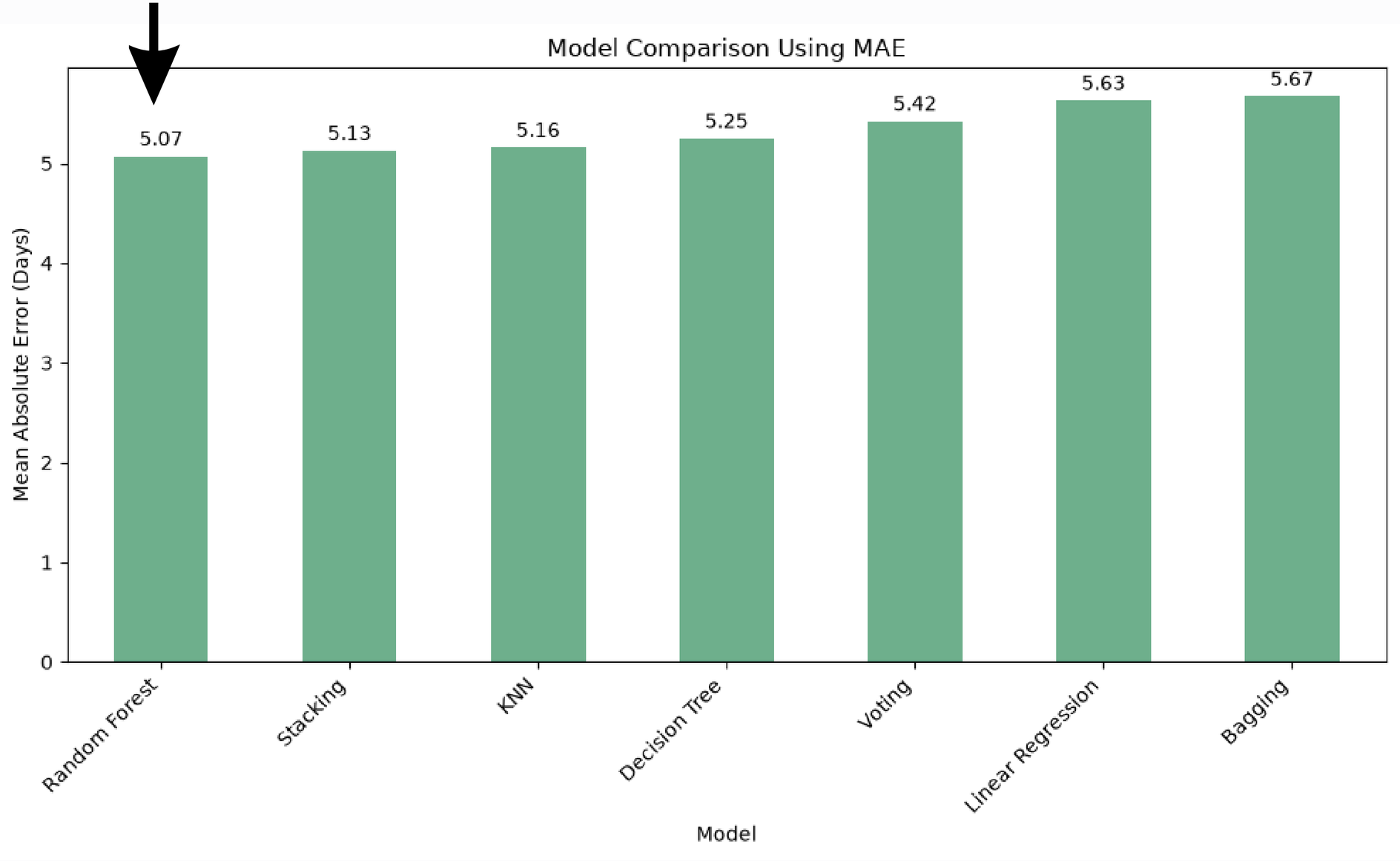
Model Evalutation

- **Easy to interpret:** Measures prediction errors in days.
- **Simple comparison:** Helps identify which model provides more accurate estimates**.**
- **Clear evaluation:** A lower MAE indicates better prediction accuracy.

Mean Absolute Error (MAE)

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} | x_i - x |$$

Final Model Selection



How Good is Our Model?

Olist Original Estimates

12.68 Days

Average Estimation Error

Our Prediction Model

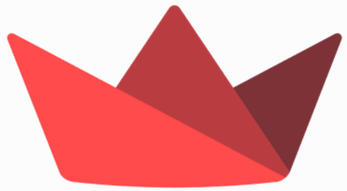
5.07 Days

Average Estimation Error

60% Improvement

Reduction in Errors

Application Demo



Streamlit

link: https://olistdelivery.streamlit.app/

Recommendations

1. **Improve Shipping Data:** Collect information about shipping companies, delivery routes, and transit times.
2. **Improve Location Accuracy:** Use actual delivery routes instead of straight-line distances.
3. **Monitor Model Performance:** Regularly compare predicted and actual delivery dates to improve accuracy.



Conclusion

In this analysis, a prediction model was developed to address the inaccuracies in Olist's delivery estimates. After preparing the historical data, multiple models were evaluated using Mean Absolute Error (MAE). The tuned Random Forest model achieved the best results, reducing the average estimation error from 12.68 to 5.07 days, approximately a 60% reduction, helping improve the accuracy and reliability of delivery estimates.

Thank You

Presented by:

Nouf Almutaweaa & Wadeaa Almutaweaa